Early Game Win Prediction in League of Legends

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Abstract

League of Legends (LoL) is a Multiplayer Online Battle Arena game (MOBA) that knows several approaches to win the game. Through teamplay or individual skill objectives can be obtained that aid in the team's victory. There is a wide range of objectives and statistics through which the game can be predicted which makes it ideal for win prediction. Several researchers have already achieved this by predicting the result of MOBA games based on pre-game, in-game or post-game data. While data from LoL is widely available, little research has been done on the game. Another MOBA game, Dota 2, shares many game mechanics with LoL and has received more attention from researchers. Algorithms such as Random Forest, Logistic Regression, k-Nearest Neighbor, Naïve Bays and Support Vector Machine have been used to create win prediction models. Most researches have focused on pre-game and post-game data. The current study addresses this by building win prediction models for in-game data. The five beforementioned algorithms were applied to LoL data to investigate whether successful models from Dota 2 are also applicable on LoL. Next to that, the applicability of the best practice model was be studied on the years from 2016 until 2021. Lastly, this thesis aimed to give insight in early objective control by distinguishing datasets and performing an feature importance analysis. The Logistic Regression model achieved best results with an accuracy of 77.99% on data at the 15 minute mark and 68.45% based on early game objectives. The application of the model on previous years yielded similar performance. Obtaining the first tower of the game was seen as the most important early game objective.

1. Introduction

In the first chapter of this thesis, an introduction is given on win prediction in League of Legends (LoL). The social and scientific relevance of the topic is discussed in section 1.1. The goal of the thesis is presented in section 1.2. The outline and main findings of the thesis are given in sections 1.3 and 1.4, respectively.

1.1 Context

E-sports are the highest form of competition in video gaming. Comparable to traditional sports, players in E-sports are required to train their mental and physical abilities to compete on the highest level (Tiedemann, 2004). The aspect in which E-sports differ from traditional sports is that E-sports competitions take place through information and communication technologies (Wagner, 2006; Witowski, 2012). This difference has frequently been a point of discussion in whether E-sports should be regarded as real sports (Dot Esports, 2021), but an increasing number of countries are recognizing professional gaming as an official sport (The Bridge, 2021). The recognition of E-sports is also supported by the large viewer numbers. For instance, approximately 46 million people watched the LoL World Championship final concurrently in 2020 (TheGamer, 2020). Moreover, Market studies on E-sports expect the viewership numbers to increase in the coming years (Newzoo, 2020; Lokhman, Karashchuk and Kornilova, 2018).

There are several genres of video games that can be characterized in E-sports competition. Multiplayer Online Battle Arena games (MOBA) such as LoL and Dota 2 are currently among the most popular considering prize pools and viewership number (Sun, 2017; Font & Mahlmann, 2018). The goal of MOBA games is to destroy the base of the enemy team. Along the way, each team member can earn gold and experience, which tends to be the best predictor of winning the game (Lee & Ramler, 2017; Cavadenti et al., 2016). In one game of LoL or Dota 2, there are also multiple objectives that a team can obtain through individual dexterity or team strategy (Donaldson, 2017). While obtaining these objectives one team can gain an advantage over their opponents, the other team naturally has access to these or other objectives as well. This results in fast-paced matches swaying back and forth in one team's favor and is explanatory for the large number of viewers attracted to E-sports (Xiao, 2021; Lee & Schoenstedt, 2011).

Both the commercial and game-related aspects of E-sports introduced a new form of sports analytics: E-sports analytics (Drachen et al., 2014). The benefits of E-sports analytics can be translated into several practical developments. For example, E-sports teams can analyze their own and their opponents' playstyle and optimize their strategy by collecting in-game data.

Next to that, bookmakers can step into the E-sports scene as well to offer fair odds through pre-match win prediction (Sweeney, Tuttle & Berg, 2021). Furthermore, game developers receive the necessary insights in order to improve the game and keep the game attractive to viewers (Lewis & Wardrip-Fruin, 2010). Researchers are keen to look in the new variant of sports analytics, as the combination of data science and E-sports analytics offers plenty of opportunities for new research.

While plenty of research has been done in the domain of result prediction in MOBA games, research on win prediction in LoL is still limited. A study of Ani, Karikumar, Devan and Deepa (2019) on win prediction in LoL showed that the Random Forest (RF) algorithm is effective, but no other algorithms have been evaluated in this game. Compared to LoL, win prediction is more thoroughly studied in Dota 2. Despite LoL and Dota 2 being different games, it is argued that data models should be applicable to other games of the same genre (Eggert, Herrlich, Smeddinck and Malaka, 2015). This statement is especially supported by the fact that most MOBA games share many key mechanics. Therefore, effective win prediction models on Dota 2 data could be applied on LoL data while obtaining similar results.

Next to the scarcity of prior research on win prediction in LoL, most research on MOBA games has focused on predicting the result of the games either before or after the game (Makarov et al., 2018; Anshori et al., 2018). Both pre-match data in the form of champion line-up and post-match data such as gold and experience earned have proven to be promising indicators of win prediction (Lee & Ramler, 2017; Cavadenti et al., 2016). The little amount of research that is done on in-game features have included variables such as kills, deaths, assists, minions slain, and gold and experience earned (Ani et al., 2019; Johansson & Wikström, 2016). Important early-game objectives such as first blood, first turret, first dragon or fist herald have not been taken into account yet. Therefore, this thesis will attempt to contribute to research on early-game win prediction revolving around objective control.

With such a diverse set of objectives in the game, different strategies have emerged over the years. Most of the strategy in LoL revolves around individual character development and cooperative teamplay in combat (Mora-Cantellops & Sicilia, 2018). Identifying patterns in strategy, individual development and combat are predictive of success (Yang, Harrison & Roberts, 2014). Therefore, the unpredictability and variability of the game must be balanced very well (Lee & Ramler, 2017). In other words, a strategy must neither be too powerful to immediately grant one team the game's victory nor too weak that the risk of gaining a certain objective does not lead to a minor advantage over the other team.

Game developers tackle this issue by updating the game occasionally, so-called balancing (Kica et al., 2016; Claypool et al., 2017). By balancing, the game remains fair but new things such as characters, items or objectives can be introduced as well. In terms of data analytics, balancing can be measured by the probability of the predicted outcome at a certain point of the game. For example, if getting the first kill in a game always results in that team winning the game, it can be said that the game is unbalanced because of its predictability. Obtaining multiple objectives in the later stages of one game, however, should make it easier to predict that that team wins the game. Logically, this usually means that the predictability in the start of a game of lower than it is towards the end of a game. Therefore, an analysis on the predictability of games over the years could say something about the development of balancing as well as the applicability of current models on future data.

1.2 *Goal of the thesis*

Following a successful RF model created for early game win prediction in Dota 2 (Johansson & Wikstrom, 2016), this research will investigate if comparable results of RF uphold for LoL. RF will be compared to a set of other algorithms as there is no study yet that indicates a best practice model for early game win prediction in LoL. The other algorithms that will be evaluated are Logistic Regression (LR), k-Nearest Neighbor (kNN), Support Vector Machine (SVM) and Naïve Bayes (NB). These algorithms have been studied in regard to MOBA games before, but have not been evaluated on either LoL or early-game data. The algorithms will be trained on a dataset with 11 early game variables at 15 minutes into the game. Besides that, a second phase of testing will be done, focusing solely on the early-game objectives consisting of 4 variables. Therefore, the first research questions will be as follows:

RQ1: To what extent can a Random Forest model predict the result of professional League of Legends matches accurately compared to Logistic Regression, k-Nearest Neighbor, Naïve Bayes and Support Vector Machine?

Since there is data available from so many years and LoL is updated regularly, it is interesting to investigate whether the developers reach their goal of keeping the game balanced. If so, the data can be used to predict winning teams based on objective control and could possibly be applied to future data. To study the effect of balancing on the predictability of professional LoL matches, the best practice model from RQ1 will be used to compare win

prediction accuracy from the datasets of 2016 until 2021. Thus, the second research question is:

RQ2: To what extent did League of Legends become a more balanced game from 2016 until 2021?

In one game of LoL, there are multiple objectives a team can obtain. As this is often a trade-off causing the other team to obtain other objectives, it can be analyzed which features have the most impact in win prediction. Getting an insight of the objectives that are most important, team strategies can be adapted to certain objectives or the game can be balanced towards them. In order to gain an understanding of how win prediction works for LoL, a sub question will be formulated that investigates feature importance. The feature importance will be extracted from the model created for RQ1. The sub question added to this research is as follows:

SQ: What role does feature importance play in win prediction of League of Legends?

1.3 *Outline*

This thesis proceeds through six chapters in order two answer both research questions. Chapter 1 has provided context for the topic of research and has introduced the research questions. Chapter 2 will zoom in on the topic of research by exploring LoL and gaining an insight of the state of current research. In chapter 3, a dataset description and an overview of the experimental set-up is given. Besides that, the candidate algorithms chosen for this thesis will be discussed. The results from the models will be presented in chapter 4. In chapter 5, the findings of this thesis will be discussed, as well as the implications, limitations and recommendations. Finally, in chapter 6, the conclusions of this thesis are presented.

1.4 *Findings*

Based on the research of Johansson and Wikström (2015), the RF algorithm was expected to perform best in win prediction of LoL matches. The results of this thesis showed that the predictive performance of RF, LR, NB and SVM is close to equal, while LR performs significantly better in computation power. In terms of balancing in the game, LoL has remained relatively equal. When considering all early game variables, predictions can be made for the

past years without much variance in accuracy. The variance increases slightly when focusing on early game objectives only.

2. Related Work

In this chapter, the key mechanics of LoL are introduced and related scientific work on win prediction is elaborated on. In section 2.1, an introduction to LoL is provided. Appendix A provides a more detailed description of the game. Section 2.3 introduces the topic of win prediction. Finally, research on win prediction in MOBA games will be explored further in section 2.4.

2.1 Introduction to League of Legends

The main game mode on LoL is a five versus five match played on a map called Summoner's Rift. One game tends to last between 20 and 40 minutes, with exceptions surpassing the 60-minute mark. Before a match starts, each player picks one of 157 characters called champions. Prior to picking their champion, each player also has the possibility to ban a champion that they do not want to play against. After the champions have been picked and banned, each player selects runes and summoner spells that provide in-game enhancements for their champion.

There are two main ways to get an advantage over the opponent in LoL. The first way is by gaining more gold and experience. Players can earn gold by last-hitting enemy minions, killing jungle monsters, destroying enemy turrets or wards, killing enemy champions and assisting enemy champion kills. Experience can be obtained by being in the vicinity of dying minions, killing jungle monsters, being nearby enemy champions kills and destroying enemy wards. It is often the case that the team that is ahead has an advantage in both gold and experience, as most ways to obtain gold and experience overlap.

Superior objective control is the second way to obtain an advantage. Objectives can be obtained in order to grant the team a boost in gold, experience, or give temporary or permanent boosts in power. Examples of these objectives are elemental dragons, elder dragons, rift heralds or baron nashors. The availability of objectives is different in three stages of the game: early game, mid game and late game. In early game, champions mainly stay in lane to generate gold and experience through minion kills and enemy champion kills. The Rift Herald, Elemental Dragons, first turrets and first enemy kill (first blood) are at stake in this phase of the game. As the first turrets have been destroyed, the game progresses into mid game. In mid game, champions slowly move out of their lanes and some team fights may occur around objectives such as Baron Nashor and the final Elemental Dragons. In the last phase, team fights occur

regularly and lanes are mostly abandoned. Baron Nashor and the Elder Dragon are the objectives to play for as they can aid a team with winning the game.

2.2 Win Prediction

There are several fields in which win prediction is a prominent tool. The political field is one of the many examples in which win prediction is used to great effect. For instance, during times of elections, it is almost impossible to turn on a news broadcast without being confronted with a prediction. One of the first win predictions based on artificial intelligence was made during the presidential elections of the United States in 1952 between Eisenhower and Stevenson (Martin, 1993). The prediction was made with the first commercial computer UNIVAC I and it predicted the outcome of Eisenhower winning with chances of 100 to 1. While scientists and the public were skeptical at first, the eventual result of Eisenhower winning put trust in the first win prediction. Nowadays, win probabilities during elections are predicted with more advanced algorithms based on more diverse datasets. Where Kumar et al. (2014) created a model to predict elections based on voter preference, accurate predictions have also been achieved with voter sentiment extracted from social media messages (Tumasjan et al., 2010; Bermingham & Smeaton, 2011).

Another field in which win prediction is applied is finance. In the field of finance, the volatile and unpredictable environment of the stock market is a great target for models of win prediction. Since there are a lot of different factors that influence the value of an index, complex models have been created to predict the future of the value. In the research of Rather, Agarwal and Sastry (2015), a linear model combined with a neural network was created in order to predict stock returns. Other researches on prediction models in finance use the combination of financial assets, liabilities, equities, revenues and expenses in order to create one summary measure. The combination of these items show the prediction as well as the probability of that prediction to happen (Phan, Sharma & Narayan, 2015; Rapach & Zhou, 2013).

Next to these formal areas, win prediction occurs in the entertainment industry as well. Recent studies made an attempt to predict the result of the Oscar Awards and the Eurovision Song Contest using sentiment analysis on Twitter (Kim, Hwang & Park, 2021; Kumulainen et al., 2020). The researches on both contests used lexicon analysis and deep learning models in order to predict the winner and the corresponding probabilities of winning for each contestant. These models and the corresponding predictions often dominate social media platforms ahead of the contests, where bookmakers provide a ranking of all contestants and where bets can be made as well.

A last area that has received a lot of attention of win prediction is sports. The range of sports in which win prediction is used is almost endless. A few examples of researches focus on basketball, American football, cricket and ice hockey (Thabtah, Zhang & Abdelhamid, 2019; Leung & Joseph, 2014; Passi & Pandey, 2018; Gu et al., 2019). The most popular algorithms that are used to build these models are SVM, NB and kNN. Besides sports, E-sports have been subject to win prediction. Games such as Rocket League, Counter Strike, Starcraft, Dota 2 and LoL have been studied frequently. In contrast to offline sports, the LR and RF algorithms are used more often, but SVM, NB and kNN are not excluded (Smithies et al., 2021; Makarov et al., 2017, Synnaeve & Bessiere, 2011; Hodge et al., 2019). Therefore, the five beforementioned algorithms will be taken into consideration for this thesis.

2.3 Win Prediction in MOBA games

Research on win prediction in MOBA games often tends to analyze one of three phases: pre-game, in-game and post-game analysis. Each of the three phases correspond to a set of algorithms that are being used most often in previous research. For pre-game analysis, kNN, NB and SVM are used most. RF and LR are frequently used in in-game and post-game analysis. The accuracy scores that these models obtain increase as the game progresses which is visible through the three different phases as well. The accuracy scores for all relevant studies are listed in Table 1. Accuracy scores for predicting the correct winning team are lowest for the models that use pre-game data and highest for the models that use post-game data. The difference in performance can be explained by the quality of the data. In-game and post-game data tell most about the progress of a game and pre-game data is often a mere indication of champion performance.

Models based on RF seem to be most promising for in-game win prediction in E-sports. Johansson and Wikström (2015) applied RF on in-game data in order to predict the winning team in Dota 2 games. Instead of sticking to predicting the winning team before, during or after the game, a minute-to-minute prediction model was made. Every time a prediction was made in a certain minute, the new values of the features were used expecting to obtain a higher accuracy during the course of the game. The model calculated its prediction using a wide set of features from kills and gold to objectives and items. Since this thesis aims to test models on their performance in the early game and on objective control, the research of Johansson and Wikström (2015) is a good starting point as it is possible to derive the accuracy at any minute of the game. However, it is uncertain if the same results will uphold for LoL data since the research of Johansson and Wikström (2015) considered data from Dota 2.

Another research proving the effectiveness of RF on in-game data is from Ani et al. (2019). Ani et al. (2019) performed an analysis on both pre-game and in-game LoL data using RF. The pre-game dataset consisted of features before the start of the game, such as champion pick and bans, and summoner spells. The in-game dataset contained features that were collected during the game but also after the game. In-game dataset features consisted of first blood, gold spent difference at the end of the game, total kills, total deaths, as well as CS difference and experience difference at ten minutes into the game. Just like the aim of this thesis, the second dataset focuses on in-game features. However, the dataset used by Ani et al. (2019) also uses post-game features and does not include objective features. Especially using the post-game features explains a very high accuracy for their model. Therefore, this research will contribute by determining whether equal results can be achieved by focusing solely on in-game data.

Besides RF, LR has also been used to a successful degree in Dota 2 researches. Eggert et al. (2015) studied the LR algorithm using post-match data as well as positional information and fighting behavior. The LR model of Eggert et al. (2015) obtained accuracy scores similar to the scores obtained in RF models. Next to that, they also included RF and NB in their research which both performed worse than their LR model. A similar research on LR was done by Makarov et al. (2018) on post-match data of Dota 2. Their accuracy score was comparable to the scores of Eggert et al. (2015), which suggests that LR is a promising algorithms for this thesis as well.

Three other relevant studies that consider pre-game data for win prediction in Dota 2. These studies mainly focused on pre-game data on the influence of character lineup instead of in-game statistics. The researchers specifically looked at the champions or heroes that were picked and banned before the game in order to estimate the winning team. The NB model from Wang and Shang (2017) performed best on the given data. The kNN model from Conley and Perry (2013) and the SVM model from Anshori et al. (2018) achieved lower accuracies based on character lineup data. The performance of pre-game models are notable worse than in-game or post-game models because of the quality of data. However, it is yet unclear how NB, kNN and SVM will perform on in-game LoL data as there is no research on this segment so far.

Table 1

Overview of the current state of research and the corresponding achieved accuracy scores.

Research	Algorithm	Data category (game)	Accuracy
Conley and Perry (2017)	kNN	Pre-game (Dota 2)	67%
Wang & Shang (2017)	NB	Pre-game (Dota 2)	85%
Anshori et al. (2018)	SVM	Pre-game (LoL)	60%
Johansson & Wikström (2015)	RF	In-game (Dota 2)	88.83%
Ani et al. (2019)	RF	In-game and post-game (Dota 2)	95.52%
Eggert et al. (2015)	LR	Post-game (Dota 2)	96.15%
Makarov et al. (2018)	LR	Post-game (Dota 2)	92%

3. Methodology

The methodological considerations are discussed in this chapter. In section 3.1, the dataset and dataset preparation is discussed. In section 3.2, the experimental set-up is discussed. The candidate algorithms are presented in section 3.3. In section 3.5, the evaluation methods are examined.

3.1 Dataset and dataset preparation

The dataset was retrieved from Oracle's Elixer. Oracle's Elixer retrieves a large number of data from professional LoL games, including the official Riot Games Application Programming Interface (API). In the datasets, comprehensive in-depth player, team and champion analytics can be found. The source is not officially endorsed by Riot Games and does not reflect their views or opinions. Nonetheless, the datasets are seen as trustworthy as they are often featured by professional broadcasts around the world or used in coverage from ESPN Esports, Dot Esports, Inven Global, theScore esports and many other outlets.

The size of the database and datasets are presented in Table 2. The amount of datapoints is significantly larger than the amount of matches, as for each match there is an individual record for each team in the match. Besides, for every game there is an single record for every individual player on each team. In total, every game in the dataset is defined by twelve datapoints. All games in both datasets are considered as highly competitive as they only include the data from the professional LoL competitions and tournaments. Matches from 2016 onwards are included because that is the year that the rift herald was introduced. If earlier years were to be incorporated in this research, no equal comparison could be made due to a different state of

the game. The features can be divided into four distinct categories: (1) metadata, (2) pre-game features, (3) in-game features, and (4) post-game features.

Table 2

An overview of the size of the datasets.

Dataset	Matches	Datapoints
2021	12273	147276
2020	9877	118524
2019	8128	97536
2018	6741	80892
2017	5704	68.448

Metadata are namespace declarations that further define certain features or datapoints within the dataset (Duval et al., 2002). The three features of metadata in the datasets were gameid, datacompleteness and url. The feature gameid contained a unique code that distinguishes every game from another. The datapoints in datacompleteness informed about the completeness of the data collected in the game.

Pre-game features include all the information that is given before the actual game starts. These features included the competition, teams and players, but also the champions picked and banned. Previous research on champion line-up mainly focused on these features of the dataset in order to predict the winning team (Eggert et al., 2015; Makarov, Savostyanov, Litvyakov & Ignatov, 2018).

In-game features are the features that are collected during one point in the game. Since most research focused on pre-game features or post-game features, in-game data will be investigated in this thesis. It was possible to divide the features in this category into two sub-categories: (1) first objective features and (2) statistical difference features. First objective features included all features about which team obtained a certain objective first. Some objectives were obtainable multiple times throughout the game, but including the total number of objectives obtained would have made the feature a post-game feature. As the first objective feature was not an indication of the total amount of objectives gained, these features could be seen as in-game features. Besides that, taking first objectives was often a trade-off of giving up another objective elsewhere on the map. Examples of early game first objective features were first blood, first dragon, first turret and first herald. Statistical difference features are features

measured during a certain point of the game. The current databases provided data at both the 10 and 15-minute mark of the game. More explicitly than first objective features, statistical difference features tended to reflect the team that is ahead more clearly. Examples of these features at their respective point in time were their difference in gold, experience, kills, assists and deaths. Information gained from earlier research on in-game features will be used in order to optimize the models in this study and will also be used as baseline models (Johansson & Wikström, 2015; Ani et al., 2019).

Post-game features are the features collected at the end of the game. These features summarized the statistical differences between the teams and also show the amount of objectives each of the teams have obtained during the entire game. Besides post-game features such as total damage, total gold, total kills, total wards, total dragons and total turrets, this category of features also showed the result of the match. In this thesis, this feature will be used as the predictive value and will also be the value on which the models will be evaluated. As previous research showed, using these post-game features to predict the winning team will achieve a very high accuracy score (Wang & Shang, 2017; Conley & Perry, 2013; Anshori et al., 2018). This result can be explained by the causality of the winning team generally having a higher amount of total gold, total damage and total objectives.

3.2 Experimental set-up

The models for this thesis are being made using the software of RStudio. Prior to the data preparation and model training, seven additional packages were installed to aid with creating the models. Appendix B gives an overview of the packages that were used besides the R base package.

Two separate datasets were created for each of every database in order to answer the two research questions. Since the focus for this research lays on the early game predictability of LoL games, the early-game features of the in-game feature category will be used. A correlation analysis was performed among the early game features in order to determine the features that were to be used for the datasets. All statistical difference features and early objective features were included in this analysis. The result of the correlation analysis is visualized in Graph 1 below. The exact values of the correlation analysis can be found in Appendix B The results of the correlation analysis were interpreted in a way that all values above 0,5 and below -0,5 were high in correlation. The correlation analysis provided the insight that all difference features were highly correlated. Next to that, the analysis showed that none of the early game objective features were high in correlation. Therefore, the choice was made

to create two different types of datasets: the 15-minute mark dataset and the early game objectives dataset. The datasets were created for all datasets from 2016 until 2021, so twelve datasets were created in total. The resemblance of features in every year made it so that the dataset preparation can be done in the same way.

The 15-minute mark datasets consisted of all features of the database that record the statistical differences between the teams fifteen minutes into the game. The early game objectives datasets consisted of all features that indicate which team took a specific objective first. The preparation for each of the datasets was performed in similar fashion. Firstly, a new dataframe was created in order to filter the player records and assure only team information was present in the set. Secondly, for both labelled datasets, the specific features belonging to the dataset were selected. An overview of all selected features is given in Table 3 below. Thirdly, the datapoints with empty cells or NA values were omitted from the datasets. Lastly, held-out train-test sets were created with 70% of the data being train data, and 30% of the data being test data.

Graph 1

Correlation analysis on all variables at 15 minutes and early game objectives.

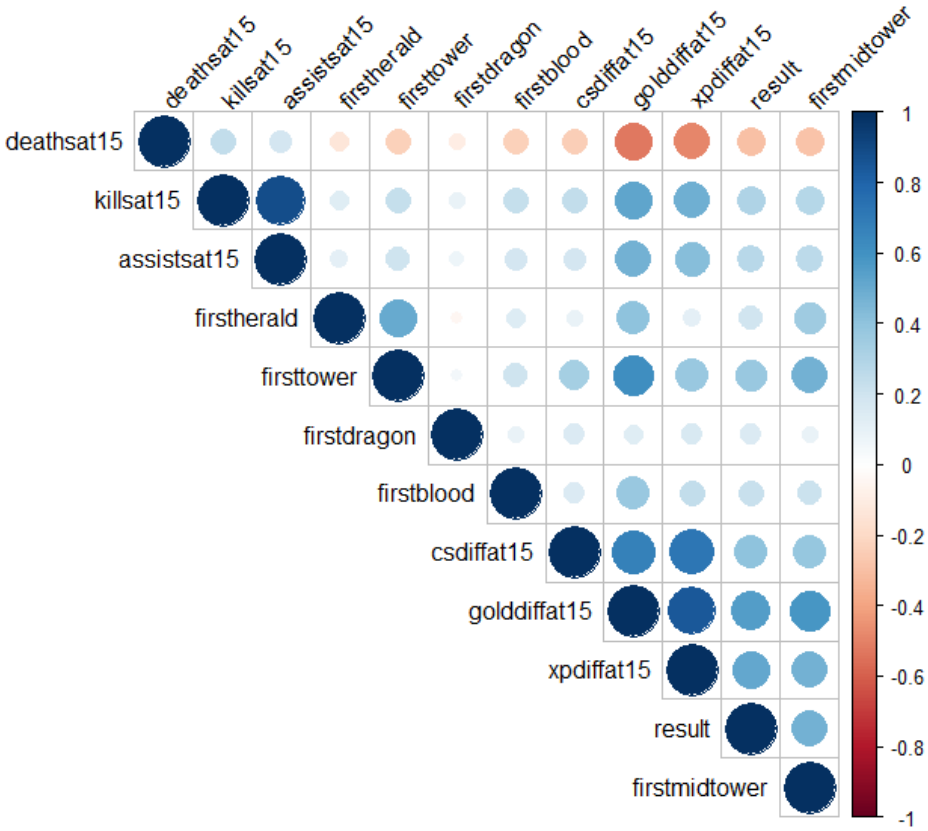


Table 1

Overview of the variables used in both datasets.

Dataset	Variables
15 minutes and early game objectives	Gold difference at 15
	Experience difference at 15
	Creep Score difference at 15
	Kills at 15
	Deaths at 15
	Assists at 15
	First mid tower
	First blood
	First dragon
	First herald
	First tower
Early game objectives	First blood
	First dragon
	First herald
	First tower

3.3 Candidate algorithms

All candidate algorithms selected for this research are explained below. Relations to previous research are made in order to support the choice of algorithm and their optimizations. In section 3.3.1, RF is discussed. LR is explored in section 3.3.2. In section 3.3.3, kNN is examined. The possibilities of SVM are explored in 3.3.4. NB is discussed in section 3.3.5.

3.3.1 Random Forest (RF)

The RF algorithm uses multiple collections of non-correlated decision trees to compute majority votes. The trees in the training set are sampled randomly without replacement. The top node of the decision tree is usually referred to as the root of the tree. From the root, the tree is recursively split at a series of decision nodes until the final decision node is reached. The final decision node is the prediction made (Kaitlin, Smith & Sandler, 2018). The algorithm is an ensemble-based learning algorithm and uses bootstrap aggregation which allows for resampling with replacement. This method reduces variance compared to other algorithms (Hastie, Tibshirani

& Friedman, 2017). The advantages of RF are that it reduces overfitting compared to regular decision trees and it works well with both categorical and continuous data. Next to that, normalization of data is not required as it uses a rule-based approach (Breiman, 2001). However, RF requires much computational power as it builds numerous trees and has to combine their outputs.

The hyperparameter tuning option that was explored for RF was mtry. The value of mtry indicates the number of variables randomly sampled as candidates at each split of nodes. There is a trade-off between a high and low value of mtry, where a low value leads to more different and less correlated trees which yields better prediction stability. However, a higher value of mtry could lead to less accuracy when RF is built on suboptimal variables (Probst, Wright & Boulesteix, 2019).

3.3.2 Logistic Regression (LR)

The LR algorithm is a probabilistic supervised learning method that associates the input to the output of the data. By solving normal equations, the algorithm finds optimal weights for the model (Zhou & Hastie, 2005). For a classification problem, LR uses a logistic function to estimate the probabilities for each outcome. No sets of features or dependencies are considered by LR among the variables and it only estimates the importance of the individual variables. Therefore, LR does not require high computation power. Another advantage of LR is that predicted parameters can give insight on feature importance. While doing this, LR can provide the positive or negative association between the variables (Eggert et al., 2015). A disadvantage is that the data must be free of missing data (Hodge et al., 2019). Therefore, the missing data values were removed from the dataset.

3.3.3 k-Nearest Neighbor (kNN)

The kNN algorithm is a non-parametric classification method. The fact that this algorithm is non-parametric makes it unique, as it is not restricted by a fixed number of parameters. The input for this algorithm consists of the number of (k) closest datapoints in the training dataset (Goodfellow et al., 2016). In case of a tie, more datapoints can be taken into consideration by hyperparameter tuning. Despite kNN being simple compared to other algorithms, the high capacity limit enables it to obtain high accuracy levels with a large dataset (Conley & Perry, 2013).

The hyperparameter option that will be optimized for the algorithm is the number of k. The value of k is the parameter that refers to the number of nearest neighbors to include in the

majority of the voting process. Generally, the smaller the number of k, the tighter the fit of the model. However, a small number of k can be noisy and influence the result. A higher number of k results in smoother decision boundaries and lower variance, but will increase bias (Conley & Perry, 2013).

3.3.4 Support Vector Machine (SVM)

The SVM algorithm is a binary classifier that finds hyperplanes that distinguish training datapoints in order to maximize generalization (Goh, Chang & Cheng, 2001). The algorithm does not provide probabilities for the output to happen, but predicts the positive and negative class identity. The algorithm works best if there is a clear margin of separation between the outcomes, if there are high dimensional spaces and where the number of dimensions is greater than the number of items. On the downside, SVM is not suitable for larger datasets and performs worse if the outcome classes overlap (Anshori et al., 2018).

Two kernels will be explored in regard of hyperparameter tuning: Radial and Polynomial. For the Radial kernel, the model will be optimized for the sigma and C values. For the Polynomial kernel, the model will be optimized for degree, scale and C. The Radial kernel tends to provide better performance but requires more computation power compared to the Polynomial kernel (Anguita et al., 2003).

3.3.5 Naïve Bayes (NB)

The NB algorithm applies the Bayes' theorem with a strong naïve assumption between the input features. The algorithm is a classifier that uses probabilities to predict the correct outcome (Goodfellow et al., 2016). First, the algorithm splits the data by labels and then models the features using the gaussian distribution. Subsequently, the posterior of each class is calculated by multiplying the likelihood of each feature occurrence together with the probability of the class. This process is repeated for every datapoint (Buckinx, Baesens, Van Den Poel & Vanthienen, 2002). NB is effective on datasets which contain both continuous and discrete data and it is especially fast making real-time predictions. Next to that, NB is insensitive to irrelevant features. A disadvantage of NB is that it assumes independent variables. Besides, the model might need re-training if new features are added to the input data (Chan, Fachrizal & Lubis, 2020).

The hyperparameter tuning options that will be optimized are fL, usekernel and adjust. Firstly, the fL parameter allows to incorporate the factor of Lacplace in case there is a problem of zero probability. Secondly, usekernel determines the kernel density estimate and is set on

continuous variables or the gaussian density estimate. Finally, the adjust parameter allows to tune the bandwidth of the kernel density. A larger value of adjust means a more flexible density estimate (Wang and Shang, 2017).

3.4 Evaluation method

The candidate algorithms were evaluated on their performance based on accuracy, F-score and computation time. Accuracy is measured by the amount of correct predictions. This means that the total correctly predicted wins and total correctly predicted losses were divided by the total amount of predictions made. The formula for accuracy is the following:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Where TP is True Positives, TN is True Negatives, FP is False Positives and FN is False Negatives. The prediction of the models will be compared with the actual results of the matches.

The F1-score is a statistic that is calculated from the precision and recall of the confusion matrices produced by the models. The F1-score is the harmonic mean of the precision and recall score and is determined between a value of 1.0 and 0. A score of 1.0 means a perfect precision and recall. The F1-scores are calculated as follows:

$$F1\text{-score} = 2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}}$$

Computation time is the last performance metric and measures the time that is needed to train or apply the model. The computation time indicates how computationally expensive it is to use an algorithm for a data-related problem. The higher the computation time, the more expensive its computation. In practice, it is often favored to strive for low computation times.

Additionally, the variables of the best practice models were be tested for overall importance. Especially regarding the early game objectives, relevant insights could be gained for team strategies. The variables that the models rely on the most, are probably the worthwhile considering in team strategy as they are responsible for a high win probability.

4. Results

In this chapter, the results of the models will be presented. The models aid in answering the two research questions and subquestion this thesis attempts to answer:

RQ1: To what extent can a Random Forest model predict the result of professional League of Legends matches accurately compared to Logistic Regression, k-Nearest Neighbor, Naïve Bayes and Support Vector Machine?

RQ2: To what extent did League of Legends become a more balanced game from 2016 until 2021?

SQ: What role does feature importance play in win prediction of League of Legends?

In section 4.1 of this chapter, the results for each of the ten models will be discussed. These results are according to the first research question. In section 4.2, the results of the best practice models from 2021 that are tested on the 2020 dataset are presented. These results are related to the second research question. The accuracies and F1 scores for all analyses can be found in Appendix E and F along with the corresponding hyperparameter values used for the models in Appendix G.

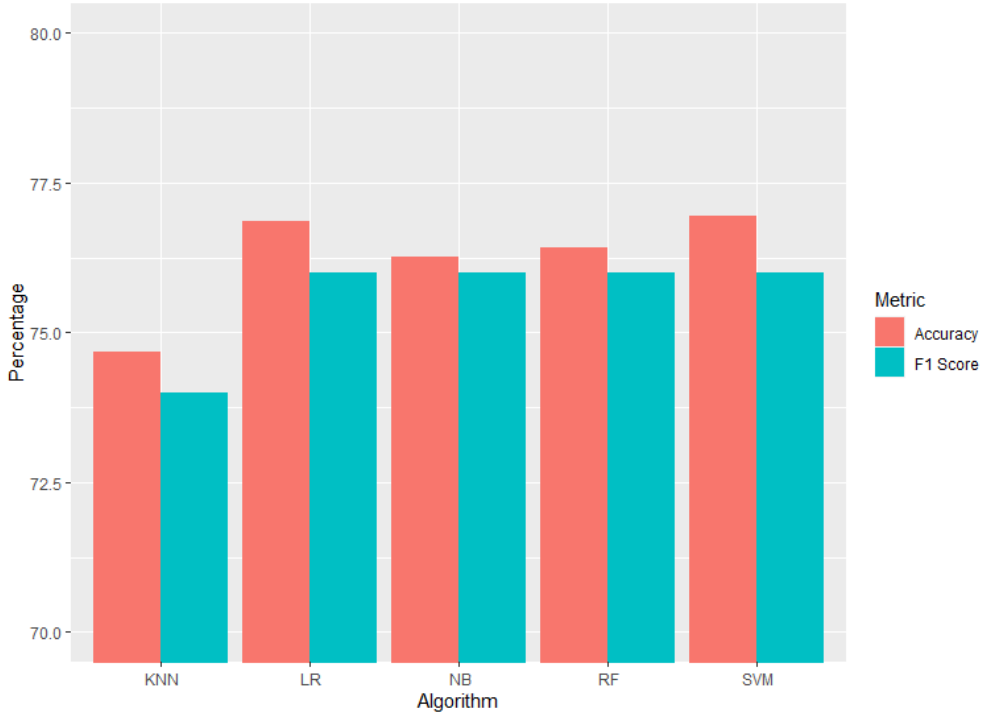
4.1 *Model performance*

Five algorithms were trained in order to create a win prediction model for two datasets of professional LoL matches in 2021. The performance of the models based on the dataset with all variables are shown in Graph 2 below. The models based on LR and SVM showed best results in regard to their accuracy score. Nonetheless, RF and NB did not predict much worse than the LR model. In fact, the F1-scores of LR, NB, RF and SVM were equal. The performance that stood out most was the result from the kNN model. Both its accuracy and F1-score showed that it performed worse than the other algorithms that were evaluated.

The performance of the five models on the objectives dataset is shown in Graph 3. The model based on kNN stood out for this dataset as well. The kNN algorithm did not manage to train on this dataset because it found to many similar patterns which caused an error in the majority vote. There were no tiebreaker parameter values that could prevent this error from happening. Therefore, the kNN obtained no accuracy score for the second dataset. The other algorithms performed almost equal to each other once again. There were minor differences in their accuracy scores but the F1-scores were similar.

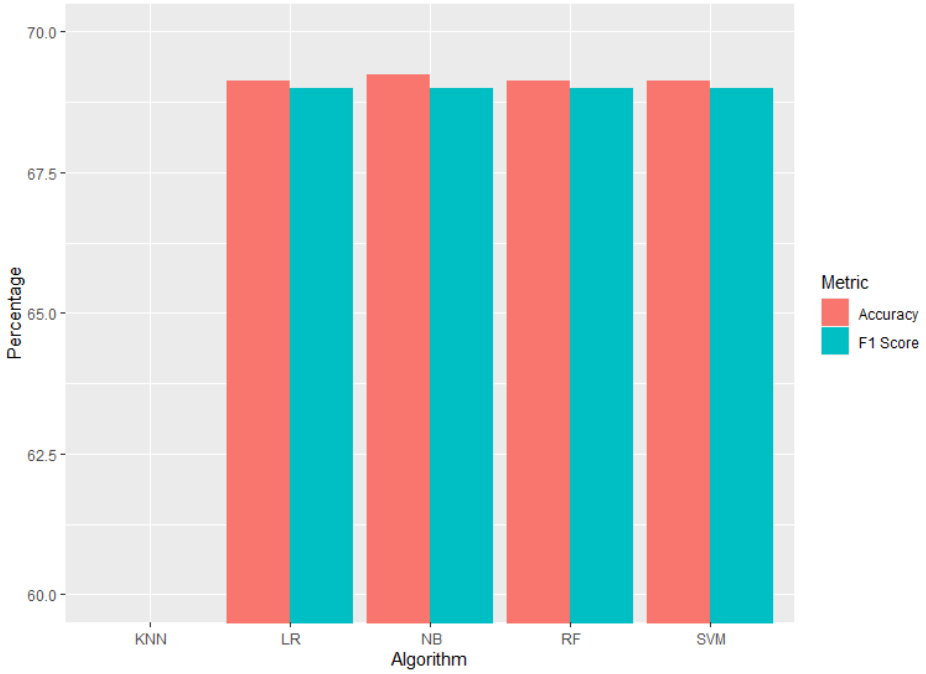
Graph 2

Evaluation of the models based on all variables at 15 minutes (limits 70% to 80%).



Graph 3

Evaluation of the models based on early game objectives (limits 60% to 70%).



The computation times for each algorithm were used as a final performance metric. The computation times for each algorithm on both datasets are provided in Table 4. The computation time for hyperparameter search and computation time were measured. LR was the algorithm which was least computationally expensive. kNN and NB showed slightly longer computation times, while RF and especially SVM took a long time computing. Despite LR, NB, RF and SVM having similar performances based on accuracy and F1-score, computation time shows LR needed less computation time. Therefore, the decision was made to apply the LR model to RQ2.

Table 4

Computation time for all models.

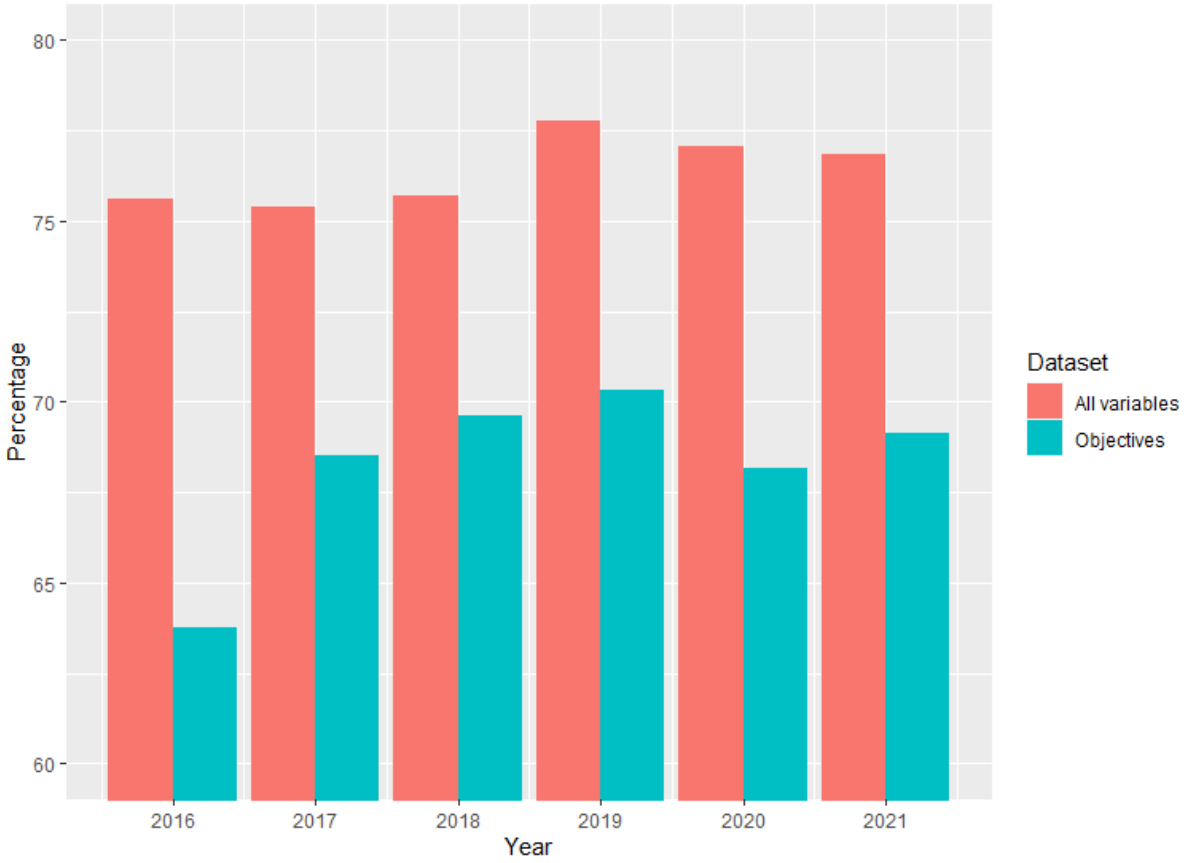
	15 minutes and early game objectives		Early game objectives	
	Hyperparameter search	Computation time	Hyperparameter search	Computation time
Random Forest	14 min 39 sec	1.17 sec	3 min 56 sec	0.55 sec
k-Nearest Neighbor	1 min 6 sec	2.69 sec	-	-
Logistic Regression	**4 sec**	**0.36 sec**	**3 sec**	**0.14 sec**
Naïve Bayes	3 min 42 sec	4.08 sec	1 min 36 sec	0.66 sec
Support Vector Machine	30 min 30 sec	3.25 sec	34 min 4 sec	3.19 sec

4.2 *Predicting game balance*

The best performing model from RQ1 was applied to data from every year since 2016 until 2021. Comparable to RQ1, a distinction was made between a dataset with all variables at 15 minutes and a dataset with early game objectives. The accuracy scores of LR on the datasets with matches from 2016 to 2021 are presented in Graph 4. The performance of LR on the datasets with all variables showed a positive or negative variance of around 1%. The model achieved slightly worse accuracy scores in the years 2016, 2017 and 2018 but performed slightly better in 2019 and 2020. The accuracy scores of LR on the dataset with early game objectives showed more variance. The positive or negative difference is 2% in the years from 2017 until 2020 compared to 2021. The model had a harder time predicting the winning team based on data from 2016 where it performed 6% worse compared to 2021.

Graph 5

Accuracy scores of the LR model on the All Variables and Early Game Objectives datasets from 2016-2021.



4.3 *Variable importance*

Lastly, an overall variable importance analysis was performed to investigate on which variables the model relies most when predicting the winning team. The variable importance analysis was only performed on the second dataset which only included the early game objectives, as these results may be the most practical for team strategies or individual gameplay. Table 5 shows an overview of the variable importance analysis.

The first indication is that LR relies heavily on the first tower taken by a team. The second takeaway is that the model relies the least on first herald taken. Lastly, the model also relies on the objectives first blood and first dragon, where it values first blood slightly higher than first dragon. These objectives are more important than the herald but less important than taking first tower.

Table 4

The overall variable importance of the early game objectives in both best practice models scaled to a score of 100.

Objective	Variable importance
First blood	51.31
First dragon	48.37
First herald	0.00
First tower	100.00

5. Discussion

This thesis attempted to evaluate different algorithms on their capability to predict the winning team in professional LoL matches. Besides that, it was investigated whether the algorithms could predict matches from 2016 until 2020 equally well as matches from 2021, and thus getting to know if the game has become more balanced. The mechanism of win prediction in LoL was further investigated by studying the variable importance of the best practice model. This chapter will discuss the findings that came from this study. In section 5.1, the key findings of the study will be reviewed and put into context with previous research. In section 5.2, the studies' implications will be discussed. The limitations of this thesis will be explained in section 5.3. In section 5.4, recommendations will be made for future research.

5.1 *Key findings*

The first key finding of this research is that the LR model was the overall best predictor of the winning team in LoL matches. LR performed equal to RF, NB and SVM based on accuracy and F1-scores but showed to be the least computationally expensive. The small amount of computational power needed can be explained by the few hyperparameters used by the algorithm. LR is a more simple algorithm which benefits in computation power but lacks in hyperparameter tuning. Nonetheless, the results obtained in this thesis contributes by proving that LR is applicable on in-game and LoL data. This supports previous research of Eggert et al. (2015) and Makarov et al. (2018) on post-game data, who showed that LR already performed well on MOBA game win prediction. Despite these results, the finding is in contrast with what was expected beforehand. Based on research of Wikström and Johansson (2015) and Ani et al. (2019), it was expected that RF would perform best as it was the only algorithm that was tested on LoL data before or in-game data in Dota 2. While the expectation was not met, RF still

showed equal results regarding its accuracy and F1-score compared with LR, despite demanding more computation power. Therefore, RF can still be seen as a possible algorithm for future research if the trade-off with computation time and hyperparameter tuning is wished to be made.

The NB and SVM algorithms that were evaluated in regard of the first research question showed new insights as well. Wang and Shang (2017) and Anshori et al. (2018) showed that NB and SVM were capable of predicting the winning team based on pre-game Dota 2 data. The NB and SVM model used in this thesis proved to be effective on in-game and LoL data as well. Especially the NB model, that outperformed LR slightly on accuracy, showed good results if it was not for the difference in computation power. The difference with LR in computation power can, however, be overseen for future use. The gap in computation power is different in comparison to SVM, for which it took significantly longer to find optimal hyperparameters and to run the model. While it is possible to tune more hyperparameters in comparison to LR and NB, a decision should be made whether this trade-off is worth making.

The kNN algorithm stood out most as it performed worse on the dataset with all variables compared to LR, RF, NB and SVM. The fact that its computation power was second fastest does not weigh up to the loss of accuracy and F1-score. Next to that, kNN was not able to predict the winning teams based on the dataset of early objectives. The inability to do so is explained by the lack of distinct patterns in the dataset. While LR, RF, NB and SVM perform well on datasets with categorical data to base its predictions on, kNN is designed to generate predictions based on numerical data. Accordingly, kNN is least suited for prediction based on data of objectives or including objectives.

The second key finding of this thesis is that LoL remained balanced in terms of win predictability. The accuracy scores from LR between 2016 and 2021 showed minor variance based on the dataset with all variables. The model showed more variance through the years when predicting games based on early game objective variables, especially in 2016. The increase in variance could be explained because important variables such as gold income and experience gained are left out in the early game objectives dataset. The drop of accuracy in 2016 could be explained by the introduction of a new objective that year: the rift herald. Most likely, the rift herald did not have much impact on the game when it was introduced as its rewards were changed in 2017. After 2017, the rewards of the rift herald remained mostly the same which explains the lesser amount of variance from that year onwards.

The last key finding concerns the variable importance found in this thesis. The variable that was found most important was the first tower while the least important variable was the rift

herald. It can be argued that the difference between the first tower and first herald exists due to the correlation between the two objectives. LoL gameplay supports this correlation because the rift herald often contributes to taking the first tower. Taking the first tower grants gold for the entire team and opens up the map for more champion movement and enemy champion vulnerability. The lesser importance of first blood is explained by the lower amount of gold that is rewarded compared to first turret, while first dragon gives only a small enhancement in early game and is becomes more valuable in late game if multiple are obtained.

5.2 *Implications*

Theoretical implications of this thesis are that it is the first study looking solely into early game objectives and the evaluation of multiple algorithms doing so. The results show that most algorithms do well in win prediction when limited to early game variables only. LR performed particularly well, closely followed by the results of NB, RF and SVM. The kNN model was the outlier with significantly lower performance. Next to that, only little variance of accuracy scores is shown over six years of professional LoL matches. This theorizes that a robust model could be made for accurate prediction and easy access for future research and win prediction in LoL. Next to that, the results of this thesis show that LR, NB, RF and SVM are suitable for in-game win prediction of LoL matches. kNN can be used when categorical data is excluded.

In practice, implications are made for win prediction, balancing and team strategy. The current study showed that LoL games can be predicted with an accuracy of around 75%. The growing e-Sports betting industry could use this information by adjusting their odds accordingly. The odds could possibly be adjusted by implementing a real-time win prediction model or adapting their odds on variable importance. Variable importance may also be of use for professional teams in planning for their matches. Strategies could be made around obtaining first tower in order to increase their chances to win the game. Lastly, the developers of LoL may use the accuracy scores provided to indicate if they are satisfied with the game state. When updating and balancing the game, developers may keep the accuracy scores and variable importance in mind and could increase or decrease them if that would suit the state of the game.

5.3 *Limitations*

There are three limitations concerning the current study on early game win predictability in LoL. The first limitation is that the datasets contained professional matches from all regions around the world. In every region, teams play according to a certain meta which means that a certain strategy or playstyle is more popular. Additionally, one region might be stronger than

the other. Therefore, the generalizability of the models can be questioned towards specific regions.

The second limitation is that the models in this thesis are solely trained on professional match data. This limitation implies that it may not be possible to use the models for casual play. Casual play often revolves more around individual skills since the means of communication are far more limited compared to competitive environments. This difference makes it harder to play around objectives and results in the focus laying more heavily on individual outplay.

The final limitation concerns the validity of the results when comparing them to research before 2020. Since 2020, certain professional regions have decided to have the teams play their matches at home instead of in the studio due to COVID-19. The change of environment may have impacted professional gameplay in strategy or individual gameplay, as the players feel less pressure from fans and have a slightly higher ping rate.

5.4 *Recommendations*

This thesis allows for several follow-up studies in the field of win prediction and prediction based on early game objectives in LoL. A first recommendation the same approach is the the analysis of the game per patch or per region to see if any differences occur in those fields. A second recommendation can be done regarding late game objectives. This thesis showed the value of early game objectives, but late game objective choice might also impact the win prediction chances of a team. As a last recommendation, the generalizability of win prediction results of different MOBAs can be further studied. While different researchers focused on different games, aspects and models, the generalizability of models between MOBA games cannot yet be proven. Research on the same aspects using the same models may give new insights in the generalizability of win prediction models in the same gaming genre.

6. Conclusion

This thesis attempted to answer a set of two research questions on win prediction in LoL. The final chapter will give an overview of how these questions have been answered.

RQ1: To what extent can a Random Forest model predict the result of professional League of Legends matches accurately compared to Logistic Regression, k-Nearest Neighbor, Naïve Bayes and Support Vector Machine?

To answer the first research question, five candidate algorithms were selected. The RF model created for this thesis proved to be a good win predictor when it comes to early game variables. LR outperformed RF by a small margin on both datasets and was least computationally expensive. Nonetheless, both LR and RF seem to be promising algorithms for LoL win prediction. NB and SVM performed comparable to LR as well but also performed worse in terms of computational power. kNN performed worst and proved to be unsuitable for categorical objective prediction.

RQ2: To what extent did League of Legends become a more balanced game from 2016 until 2021?

To answer the second research question, LR was taken and applied on new datasets of matches from 2016 until 2020. The results from the models showed that there was only a small variance in accuracy. Despite the many changes LoL has had over the years, it could be argued that the game has remained the same in regard of balancing and predictability. Next to that, it showed that the models could be used for future prediction as the evaluation metrics show little variance over the past years.

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Appendix

Appendix A – Description of League of Legends

League of Legends is a MOBA game developed by Riot Games (Donaldson, 2017). The game was released in 2009 for Microsoft Windows and in 2013 for macOS. League of Legends was originally inspired by the predecessor of Dota 2, Defense of the Ancients (Dota), and is basically a simplified and more accessible version of Dota. After League of Legends' release in 2009, the game counted 302,000 unique monthly players on average. In 2021, this number has risen to 104,000,000 unique monthly players. The game is free-to-play and is monetized through purchasable character customization. Character customization does not mean that League of Legends is a pay-to-win game, since the purchases are exclusively cosmetic and do not provide in-game powers.

The main game mode on League of Legends is a five versus five match played on a map called Summoner's Rift. One game tends to last between 20 and 40 minutes, with exceptions surpassing the 60-minute mark. Before a match starts, each player picks one of 157 characters called 'champions'. Every champion has a unique set of abilities and strengths. Prior to picking their champion, each player also has the possibility to ban a champion that they do not want to play against. After the champions have been picked and banned, each player can select runes that provide in-game enhancements for their champion according to the player's desired playstyle. Lastly, a set of summoner spells have to be selected, which are additional abilities for the champion.

Once the game has loaded, the champions spawn at their nexus on Summoner's Rift and are able to move around the map. Summoner's Rift is a square map on which one team defends the blue side and the other defends the red side. The blue side nexus is located at the bottom left corner and the red side nexus at the upper right corner. This is also the place where champions respawn after a short delay when they are killed. From base to base along the top border of the map runs the top lane and along the bottom border runs the bot lane. Diagonally from base to base runs the mid lane. Diagonally from the upper left to lower right flows a passable river, that splits the blue side of the map from the red side of the map. In between the lanes is an area called the jungle.

During the game, non-playable characters (NPCs) called minions spawn from both bases and move through every lane in the form of waves. When ignored, these minions collide in the middle of each lane. In the jungle, neutral monsters can be found that do not fight for any side, but will defend themselves once attacked. Slaying minions and monsters are the main income source of gold and experience for champions. Minions are also referred to as creeps,

and the total amount of minions slain is referred to as creep score or CS. Turrets can be destroyed for gold as well. Each lane has three turrets which a team needs to destroy in order for them to reach the opponent's base. The base counts three inhibitors, one for every lane, which have to be destroyed to be able to advance to the two nexus turrets guarding the nexus. Destroying an inhibitor also grants a buff to the minions in that specific lane. The team that manages to destroy the enemy's nexus first wins the game.

There are two main ways to get an edge over the opponent in a game of League of Legends. The first way to get ahead is by gaining more gold and experience than the opponent. Players can earn gold by last-hitting enemy minions, killing jungle monsters, destroying enemy turrets or wards, killing enemy champions and assisting enemy champion kills. Players can buy items at their base with the gold they earned. These items give them a significant boost in power. Next to earning gold, experience can be obtained by being in the vicinity of dying minions, killing jungle monsters, as well as being nearby enemy champions being killed and destroying enemy wards. Experience causes champions to level up to a maximum of level 18. Gaining a level grants champions access to new or stronger abilities and strengthens the champion. It is often the case that the team that is ahead has an advantage in both gold and experience, as most of the ways to obtain them overlap. The gold and experience earned in one game is, however, discrete and does not carry over to other games.

The second way to obtain an advantage over the opponent is to have better objective control. Several of these objectives can be obtained in order to grant the team a boost in gold, experience, or give temporary or permanent boosts in power. Examples of these objectives are elemental dragons, elder dragons, rift heralds or baron nashors. The availability of objectives and the state of the game can be divided into three phases: early game, mid game and late game. In early game, champions mainly stay in lane to generate gold and experience through minion kills and some enemy champion kills. The Rift Herald, Elemental Dragons, first turrets and first enemy kill (first blood) are at stake in this phase of the game. As the first turrets have fallen, the game progresses into mid game. In mid game, champions slowly move out of lane and some teamfights may occur objectives. These objectives are Baron Nashor and the last of four Elemental Dragons for one team. Sometimes games end after mid game because the lead has already grown significantly. Most of the times, however, the game advances into late game. In the last phase, teamfights occur regularly and lanes are mostly abandoned. Baron Nashor and the Elder Dragon are the objectives to play for as they can aid a team with finishing the game.

The competitive scene of League of Legends is widespread across the world. The scene counts as much as twelve major regions in 2021. Four of these regions have franchised systems, namely Korea, Europe, China and North America. Most of these regions have a competition of eight to twenty teams which play each other in a double-round robin. Each team consists of five top-players which are contracted by the team and may earn similar wages compared to professional athletes. Usually, regional competitions are held in spring and summer, called the Spring Split and Summer Split. Following the double-round robin format, play-offs will be played among the top teams to determine the winner of that split. Teams that do well in regional competitions and manage to reach far into the play-off tournaments, can qualify for international tournaments. The two most prestigious tournaments are the Mid-Season Invitational and the World Championship. The Mid-Season Invitational takes place between the Spring and Summer Split and only invites the top seeds from every region in Spring Split. The World Championship takes place in fall after the Summer Split. Each of the four major regions is represented by three teams, while the minor regions can either send one or two teams to the tournament.

Appendix B – An overview of the additional R packages used besides the R base package.

Package	**Use**
ggplot2	Aids with creating graphics based on the data provided.
dplyr	Enhances possibilities for data manipulation
caret	Gives access to the process of creating the predictive models.
e1071	Gives access to the process of creating the predictive models.
randomForest	Grants additional tools for the randomForest model.
corrplot	Allows the use of a function to create a correlation plot.
kernlab	Grants additional kernel-related tools for the Support Vector Machine models.

Appendix C - Overview of all variables in the database

gameid	The unique code that belongs to the match
datacompleteness	States if the datapoint has no missing items
url	The url linking to the online match overview
league	The competition in which the match was played
year	The year in which the match was played
split	The split in which the match was played
playoffs	Indicated if the match was played in playoffs
date	The date on which the match was played
game	The game number if the match happens to be a series
patch	The patch on which the game was played
playerid	The unique code that belongs to a player in the match
side	The side that the team plays on
position	The role that the player has
player	The in-game name of the player
team	The team that competes in the match
champion	The champion that is played
ban1	The first banned champion of the team
ban2	The second banned champion of the team
ban3	The third banned champion of the team
ban4	The fourth banned champion of the team
ban5	The fifth banned champion of the team
gamelength	The length of the match
result	The result of the match
kills	The total amount of kills at the end of the match
deaths	The total amount of deaths at the end of the match
assists	The total amount of assists at the end of the match
teamkills	The teams' total amount of kills at the end of the match
teamdeaths	The teams' total amount of deaths at the end of the match
doublekills	The total amount of doublekills
triplekills	The total amount of triplekills
quadrakills	The total amount of quadrakills
pentakills	The total amount of pentakills

firstblood	The team or player who got the first kill
firstbloodkill	The player who got the first kill
firstbloddassist	The player(s) who got an assist on the first kill
firstbloodvictim	The player that died first
teamkpm	The team kills per minute
ckpm	Average combined kills per minute
firstdragon	The team that got first dragon
dragons	The total amount of dragons slain
opp_dragons	The total amount of dragons slain by the opponent
elementaldragons	The total amount of elemental dragons slain
opp_elementaldragons	The total amount of elemental dragons slain by the opponent
infernals	The total amount of infernal dragons slain
mountains	The total amount of mountain dragons slain
clouds	The total amount of cloud dragons slain
oceans	The total amount of ocean dragons slain
dragons (type unkown)	The total amount of dragons slain of which the type is unknown
elders	The total amount of elder dragons slain
opp_elders	The total amount of elder dragons slain by the opponent
heralds	The total amount of heralds slain
opp_heralds	The total amount of heralds slain by the opponent
firstbaron	The team that got the first baron
barons	The total amount of barons slain
opp_baron	The total amount of barons slain by the opponent
firsttower	The team that destroyed the first tower
towers	The total amount of towers destroyed
opp_towers	The total amount of towers destroyed by the opponent
firstmidtower	The team that destroyed the first tower in mid lane
firsttothreetowers	The team that destroyed three towers first
turretplates	The total amount of turret plates destroyed
opp_turretplates	The total amount of turret plates destroyed by the opponent
inhibitors	The total amount of inhibitors destroyed
damagetochampions	The total amount of damage dealt to champions
dpm	Damage to champions per minute

damageshare	Percentage of damage to champions dealt of total damage to champions dealt
damagetakenperminute	Damage taken per minute
damagemitigatedperminute	Damage mitigated per minute
wardsplaced	The total amount of wards placed
wpm	Wards placed per minute
wardskilled	The total amount of wards killed
wcpm	Wards cleared per minute
controlwardsbought	The total amount of control wards bought
visionscore	Vision score
vspm	Vision score per minute
totalgold	Total gold earned at the end of the game
earnedgold	Total gold earned deducted by passive income
earned gpm	Total gold earned per minute
earnedgoldshare	The percentage of gold earned compared to total earned gold
goldspent	Total amount of gold spent
gspd	Average gold spent difference
total cs	Total amount of Creep Score
minionkills	Total amount of minion kills
monsterkills	Total amount of monster kills
monsterkillsownjungle	Total amount of monster kills in the own jungle
monsterkillsenemyjungle	Total amount of monster kills in the enemy jungle
cspm	Average Creep Score per minute
goldat10	Gold earned at 10 minutes
xpat10	Experience earned 10 minutes
csat10	Creep Score 10 minutes
opp_goldat10	Opponent gold earned 10 minutes
opp_xpat10	Opponent experience earned 10 minutes
golddiffat10	Gold difference at 10 minutes
xpdiffat10	Experience difference at 10 minutes
csdiffat10	Creep Score difference at 10 minutes
killsat10	Kills at 10 minutes
assistsat10	Assists at 10 minutes

deathsat10	Deaths at 10 minutes
opp_killsat10	Opponent kills at 10 minutes
opp_assistsat10	Opponent assists at 10 minutes
opp_deathsat10	Opponent deaths at 10 minutes
goldat15	Gold earned at 15 minutes
xpat15	Experience earned 15 minutes
csat15	Creep Score 15 minutes
opp_goldat15	Opponent gold earned 15 minutes
opp_xpat15	Opponent experience earned 15 minutes
golddiffat15	Gold difference at 15 minutes
xpdiffat15	Experience difference at 15 minutes
csdiffat15	Creep Score difference at 15 minutes
killsat15	Kills at 15 minutes
assistsat15	Assists at 15 minutes
deathsat15	Deaths at 15 minutes
opp_killsat15	Opponent kills at 15 minutes
opp_assistsat15	Opponent assists at 15 minutes
opp_deathsat15	Opponent deaths at 15 minutes

Appendix D - Correlation matrix of all variables at 15 minutes and early game objectives

	result	firstblood	firstdragon	firstherald	firsttower	firstmidtower
result	1.00	0.22	0.16	0.20	0.37	0.48
firstblood	0.22	1.00	0.09	0.14	0.20	0.21
firstdragon	0.16	0.09	1.00	-0.05	0.05	0.08
firstherald	0.20	0.14	-0.05	1.00	0.50	0.35
firsttower	0.37	0.21	0.05	0.50	1.00	0.48
firstmidtower	0.48	0.21	0.10	0.35	0.48	1.00
golddiffat15	0.56	0.38	0.13	0.40	0.61	0.58
xpdiffat15	0.51	0.24	0.17	0.11	0.38	0.47
csdiffat15	0.40	0.16	0.16	0.09	0.34	0.39
killsat15	0.30	0.23	0.09	0.14	0.23	0.29
assistsat15	0.28	0.19	0.08	0.11	0.20	0.26
deathsat15	-0.30	-0.23	-0.09	-0.14	-0.23	-0.29

	golddiffat15	xpdiffat15	csdiffat15	killsat15	assistsat15	deathsat15
result	0.56	0.51	0.40	0.30	0.28	-0.30
firstblood	0.38	0.24	0.16	0.23	0.19	-0.23
firstdragon	0.13	0.17	0.16	0.09	0.08	-0.09
firstherald	0.40	0.11	0.09	0.14	0.11	-0.14
firsttower	0.61	0.38	0.34	0.23	0.20	-0.23
firstmidtower	0.58	0.47	0.39	0.29	0.26	-0.28
golddiffat15	1.00	0.84	0.67	0.52	0.47	-0.52
xpdiffat15	0.84	1.00	0.73	0.49	0.43	-0.49
csdiffat15	0.67	0.73	1.00	0.25	0.18	-0.25
killsat15	0.52	0.49	0.25	1.00	0.88	0.24
assistsat15	0.47	0.43	0.18	0.88	1.00	0.19
deathsat15	-0.52	-0.49	-0.25	0.24	0.19	1.00

Appendix E - The accuracy scores of the models on the 2021 matches dataset.

	15 minutes and early game objectives	Early game objectives	Average
Random Forest	77.26%	68.45%	72.86%
k-Nearest Neighbor	74.83%	-	-
Logistic Regression	77.99%	68.45%	73.22%
Naïve Bayes	76.82%	68.16%	72.50%
Support Vector Machine	76.99%	68.16%	72.58%
Average	76.78%	68.31%	

Appendix F - The accuracy scores of the models predicting wins in the matches from 2016 until 2021.

	15 minutes and early game objectives	Early game objectives
2021	76.86%	69.14%
2020	77.06%	68.18%
2019	77.79%	70.33%
2018	75.70%	69.61%
2017	75.39%	68.51%
2016	75.61%	63.78%

Appendix G - Hyperparameter values used for all models.

	15 minutes and early game objectives	Early game objectives
RF	mtry = 2	mtry = 2
LR	-	-
kNN	k = 9	-
NB	fL = 0 usekernel = TRUE adjust = 1	fL = 0 usekernel = FALSE adjust = 1
SVM	Kernel = Radial Sigma = 0.711 C = 1	Kernel = Radial Sigma = 0.167 C = 0.25